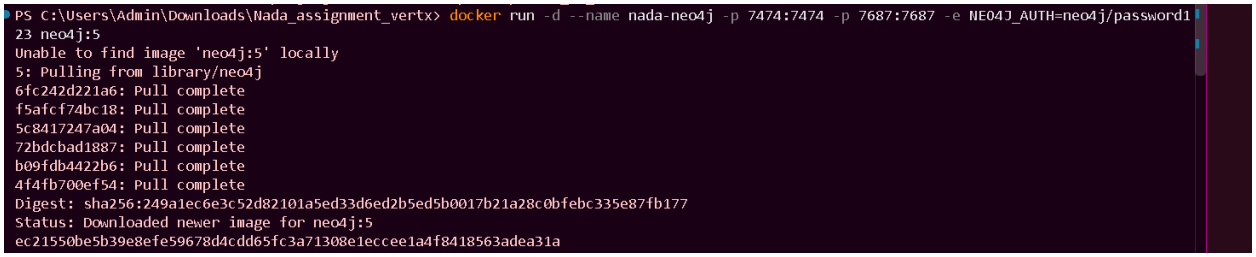
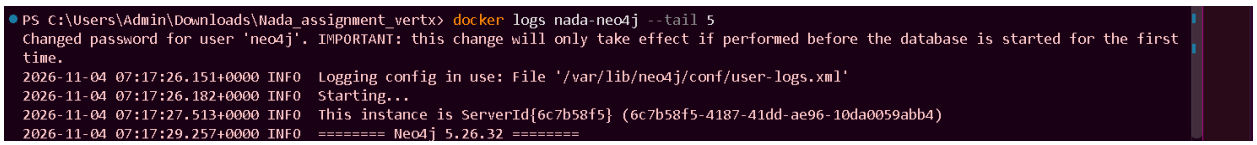
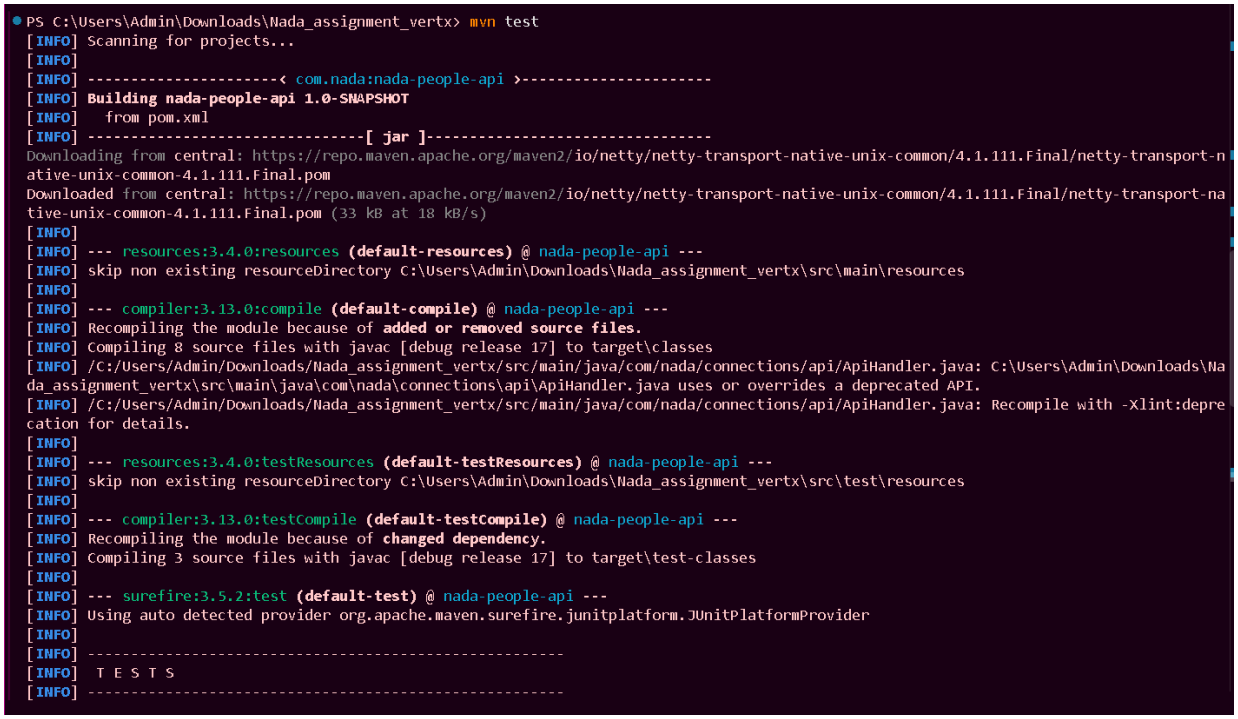
1.Pulling Neo4j. Started Neo4j 5 in Docker on my Windows laptop with `docker run`. The image wasn't cached, so it was pulled first.



2. Neo4j starting. `docker logs nada-neo4j` shows Neo4j 5.26 starting up locally. Once it's up, the Bolt port 7687 is ready for the app.



3. Maven build. Running `mvn test` in the VS Code terminal on my laptop. Maven downloads the dependencies (Vert.x, the Neo4j driver) and compiles the sources and tests.



4. Tests pass, server starts. All 65 tests pass (`BUILD SUCCESS`, about 20 s). The tests use the in-memory store, so they don't need Neo4j. Then I set `NEO4J_PASSWORD` and ran `mvn compile exec:java`. The app connects to `bolt://localhost:7687` and the Vert.x server starts on port 8080

```
PS C:\Users\Admin\Downloads\Nada_assignment_vertx> mvn test
[INFO] Tests run: 22, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.075 s -- in com.nada.connections.service.PeopleServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 65, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  20.420 s
[INFO] Finished at: 2026-11-04T12:48:10+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Admin\Downloads\Nada_assignment_vertx> $env:NEO4J_PASSWORD="password123"
PS C:\Users\Admin\Downloads\Nada_assignment_vertx> mvn compile exec:java
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.nada:nada-people-api >----------------------
[INFO] Building nada-people-api 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ nada-people-api ---
[INFO] skip non existing resourceDirectory C:\Users\Admin\Downloads\Nada_assignment_vertx\src\main\resources
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ nada-people-api ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- exec:3.3.0:java (default-cli) @ nada-people-api ---
Using Neo4j at bolt://localhost:7687
Server running on http://localhost:8080
```

5. Creating people. Using `Invoke-RestMethod` to `POST /people` for Asha, Ravi and Meera. Each person gets a random 32-bit id written as 8 hex characters (Asha's is `9f7d287b`), not 1, 2, 3.

```
PS C:\Users\Admin\Downloads\Nada_assignment_vertx> $a = Invoke-RestMethod -Method Post -Uri http://localhost:8080/people -ContentType "app
lication/json" -Body '{"name":"Asha"}'
PS C:\Users\Admin\Downloads\Nada_assignment_vertx> $b = Invoke-RestMethod -Method Post -Uri http://localhost:8080/people -ContentType "app
lication/json" -Body '{"name":"Ravi"}'
PS C:\Users\Admin\Downloads\Nada_assignment_vertx> $c = Invoke-RestMethod -Method Post -Uri http://localhost:8080/people -ContentType "app
lication/json" -Body '{"name":"Meera"}'
PS C:\Users\Admin\Downloads\Nada_assignment_vertx> $a

id       name
--       ----
9f7d287b Asha
```

6. Connect, contacts and path. I connected Asha–Ravi and Ravi–Meera with `POST /knows`. Asha's contacts show only Ravi, her direct connection. `GET /path` from Asha to Meera returns `connected: True`, `hops: 2`, with Ravi in the middle.

```
PS C:\Users\Admin\Downloads\Nada_assignment_vertx> Invoke-RestMethod -Method Post -Uri http://localhost:8080/knows -ContentType "applicati
on/json" -Body (@{a=$a.id;b=$b.id} | ConvertTo-Json)

PS C:\Users\Admin\Downloads\Nada_assignment_vertx> Invoke-RestMethod -Method Post -Uri http://localhost:8080/knows -ContentType "applicati
on/json" -Body (@{a=$b.id;b=$c.id} | ConvertTo-Json)

PS C:\Users\Admin\Downloads\Nada_assignment_vertx> Invoke-RestMethod "http://localhost:8080/people/$($a.id)/contacts"

id       name
--       ----
b5094e7a Ravi


PS C:\Users\Admin\Downloads\Nada_assignment_vertx> Invoke-RestMethod "http://localhost:8080/path?from=$($a.id)&to=$($c.id)"

connected hops path
--------- ---- ----
     True    2 {@{id=9f7d287b; name=Asha}, @{id=b5094e7a; name=Ravi}, @{id=a0f...
```

7. Unknown person. A path request with a well-formed id that doesn't exist (`ffffffff`) returns 404 with `{"error":"no person ffffffff"}`. PowerShell shows it as red error text, which is expected.

```
PS C:\Users\Admin\Downloads\Nada_assignment_vertx> Invoke-RestMethod "http://localhost:8080/path?from=$($a.id)&to=ffffffff"
Invoke-RestMethod : {"error":"no person ffffffff"}
At line:1 char:1
+ Invoke-RestMethod "http://localhost:8080/path?from=$($a.id)&to=ffffff ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest:HttpW
   ebRequest) [Invoke-RestMethod], WebException
    + FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell
   .Commands.InvokeRestMethodCommand
```

8. Graph in Neo4j. In the Neo4j Browser at `localhost:7474`, `MATCH (p:Person) RETURN p` shows the three people stored as `Person` nodes with their random ids.

